PIPING AND INSTRUMENT DIAGRAM STP SYSTEM

PROJECT :

PT. MULIA BOGA RAYA Tbk.

PT. GRINVIRO BIOTEKNO INDONESIA

Water and Waste Water Treatment System

2025

DOCUMENT REVISION RECORD

DATE	REVISION	DESCRIPTION
23/09/2025	00	- P&ID STP 30 CMD First Issued
28/01/2026	01	- Addition of a partition wall between the aeration tank (TK-02) and the MBR tank (MBR-01). - Addition of an air lift system from MBR-01 to TK-02 (3/4" air pipe and 3" air lift pipe). - Prime tank for the backwash system is omitted due to positive suction conditions. - Backwash drain line to the equalizing tank is omitted; discharge is routed via the sampling valve with an additional elbow. - Modification of TP-01 suction piping: 2" suction line, 1 1/2" the discharge line.

PROJECT NAME

PROCHIZ GARUDAFOOD RANCAEKEK
PROJECT
STP 30 CMD

ACKNOWLEDGE BY:

ANDI BAMBANG EKO SUSILO
PROJECT LEADER

EBIT TRIYOGO
ENGINEER

ACKNOWLEDGE BY:

BESLINTON PARDEDE
CONSTRUCTION MANAGER

R JUNI INDERAWAN
ENGINEER

P&ID

No.	REVISION	DATE

SUBMITTED BY:
CONTRACTOR

PT. GRINVIRO BIOTEKNO INDONESIA
Water and Waste Water Treatment System

AGUNG WIBOWO
Project Manager

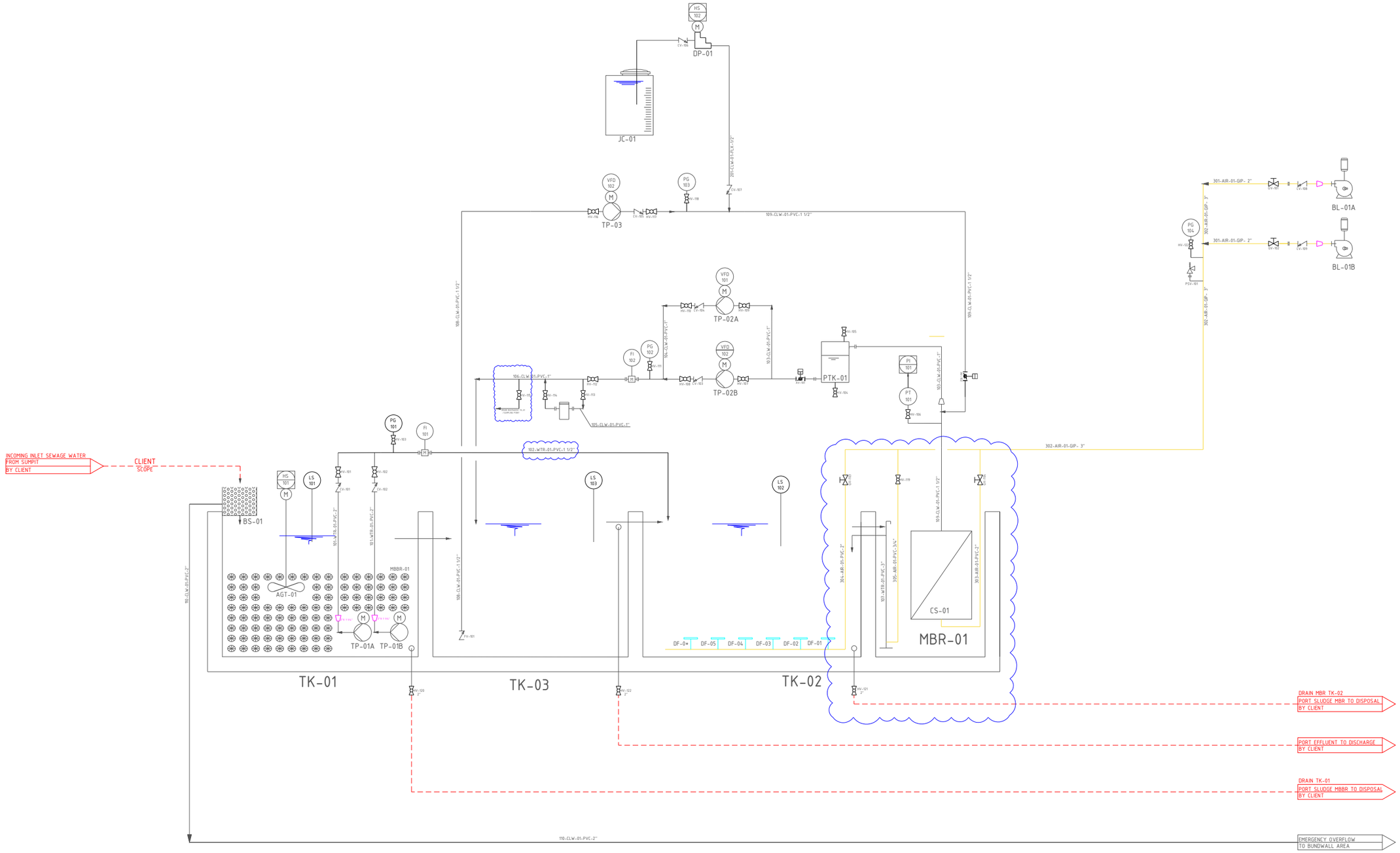
PERSON RESPONSIBLE	NAME	SIGN
DRAWN BY	ERH	
CHECK BY	NAR	

DRAWING TITLE :

PIPING AND INSTRUMENT DIAGRAM
REVISION LIST

DWG SCALE	NO SCALE
DWG REFERENCE	

DWG NUMBER



TAG.NO	TP-01A/B
EQUIPMENT	ANOXIC TRANSFER PUMP
TYPE.	SUBMERSIBLE PUMP
CAPACITY	6 CMH HEAD 7 m
POWER	0.25 kW/1 PHASE/220 V
QTY.	2 UNITS (1W+1S)

TAG.NO	TP-02A/B
EQUIPMENT	MBR PERMEATE PUMP
TYPE.	VERTICAL MULTISTAGE PUMP
CAPACITY	1.5 CMH HEAD 12-16 m
POWER	0.37 kW/3 PHASE/380 V
QTY.	2 UNITS (1W+1S)

TAG.NO	TP-03
EQUIPMENT	MBR BACKWASH PUMP
TYPE.	VERTICAL MULTISTAGE PUMP
CAPACITY	4 CMH HEAD 20-22 m
POWER	0.55 kW/3 PHASE/380 V
QTY.	1 UNIT

TAG.NO	BL-01A/B
EQUIPMENT	MBR BLOWER
TYPE.	SIDE CHANNEL BLOWER
CAPACITY	150 m3/h, 260 mbar
POWER	1.5 kW
QTY.	2 UNITS (1W+1S)

TAG.NO	TK-01
EQUIPMENT	ANOXIC TANK
VOLUME	10 m3
MATERIAL	MILD STEEL
QTY.	1 UNIT

TAG.NO	TK-02
EQUIPMENT	AERATION-MBR TANK
TOTAL VOLUME	11,7 m3
MATERIAL	MILD STEEL
QTY.	1 UNIT

TAG.NO	TK-03
EQUIPMENT	PERMEATE TANK
VOLUME	3,75 m3
MATERIAL	MILD STEEL
QTY.	1 UNIT

TAG.NO	DP-01
EQUIPMENT	CEB DOSING PUMP
CAPACITY	50 LPH ; HEAD 2 BAR
POWER	0.06 kW/1 PHASE
QTY.	1 UNIT

TAG.NO	BS-01
EQUIPMENT	BASKET SCREEN
QTY.	1 UNIT

TAG.NO	MBBR-01
EQUIPMENT	MOVING BED BIOFILM REACTOR
QTY.	1 LOT

TAG.NO	AGT-01
EQUIPMENT	AGITATOR MIXER
POWER	0.55 kW
QTY.	1 UNIT

TAG.NO	PTK-01
EQUIPMENT	PRIME TANK
VOLUME	MIN. 5-6 L
QTY.	2 UNIT

TAG.NO	DF-0*
EQUIPMENT	DIFFUSER AERATION
QTY.	14 pcs

TAG.NO	JC-01
EQUIPMENT	JERICEN TANK
CAPACITY	25 L
QTY.	1 UNIT

TAG.NO	CLT-01
EQUIPMENT	CHLORINE TABLET
QTY.	1 UNIT

TAG.NO	CS-01
EQUIPMENT	CASSETE MEMBRANE
TYPE	PVDF

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GBI-25-MBR-STP-PID.001

LEGENDS

VALVE SYMBOLS

- BUTTERFLY VALVE
- FLEXIBLE JOINT
- CHECK VALVE
- BALL VALVE
- AUTOMATIC VALVE
- BUTTERFLY SELENOID VALVE
- FOOT VALVE
- PRESSURE SAVETY VALVE
- BREATH VALVE
- LOOSE JOINT
- AUTOMATIC EXHAUST VALVE

ABBREVIATIONS INSTRUMENT & CONTROLS

AE	ANALYZER ELEMENT
AIT	ANALYZER TRANSMITTER
AIC	ANALUZER INDICATOR CONTROL
FIT	FLOW INDICATOR TRANSMITTER
FQI	FLOW QUANTITY INDICATOR
H/O/A	HAND OFF AUTO SWITCH
HS	HAND SWITCH
LG	LEVEL GAUGE
LT	LEVEL TRANSMITTER
LIC	LEVEL INDICATOR CONTROL
LS	LEVEL SWITCH
MEI	MAN EQUIPMENT INDICATOR
PG	PRESSURE GAUGE
PT	PRESSURE TRANSMITTER
PIC	PRESSURE INDICATOR CONTROL
SIC	SPEED INDICATOR CONTROL
VFD	VARIABLE FREQUENCY DRIVE

BALLON INSTRUMENTS

- FIELD MOUNTED
- NORMALLY ACCESSIBLE TO AN OPERATOR

- FIELD OR LOCAL CONTROL PANEL
- NORMALLY ACCESSIBLE TO AN OPERATOR

- CENTRAL OR MAIN CONTROL ROOM
- NOT NORMALLY ACCESSIBLE TO AN OPERATOR

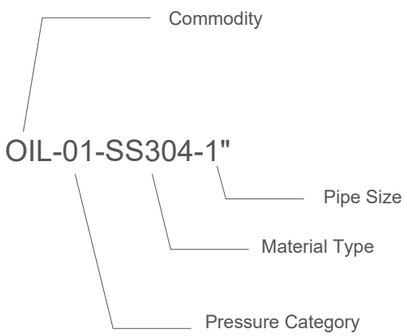
- BASIC PROCESS CONTROL SYSTEM (BPCS)
- VISIBLE IN FIELD OR FIELDS MOUNTED

- BASIC PROCESS CONTROL SYSTEM (BPCS)
- DISPLAY IN CENTRAL OR MAIN CONTROL ROOM

EQUIPMENT SYMBOLS

- M — ELECTROMAGNETIC FLOWMETER
- GENERAL BLOWER
- GENERAL PUMP
- DOSING PUMP
- ROTARY DRUM SCREEN
- SUBMERSIBLE MIXER
- VERTICAL MIXER AGITATOR

PIPE TAGGING



NOTE FOR PIPE TAGGING:

1. Line Number
   - 10* = Main Interconnection Pipe
   - 20* = Chemical Line Pipe
   - 30* = Air/Biogas Line Pipe
   - * Line Pipe Sequences

2. Commodity
   - WTR = Water Reject
   - SLG = Sludge
   - BGS = Biogas
   - AIR = Air
   - OIL= Oil
   - CLW = WATER

3. Pipe Pressure Category
   - 01 = 0-5 bar
   - 02 = >5-10 bar
   - 03 = >10-15 bar

4. Material Type
   - HDPE = HDPE Pipe
   - SS304 = SS304 Pipe
   - PVC = PVC Pipe
   - SS316L = SS316L Pipe

5. Pipe Size
   - 1" , 11/2", 2", 21/2", 3" dst

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